CS 270 – Lab 9 Q3

Question 3: (Q a b)

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;input-contract: a and b are positive integers
;output-contract: (Q a b) is a recursively defined integer
;                  -- pattern redacted for students
(define (Q a b)
  (if (= b 1) a (+ a (Q a (- b 1)))))
```

Use Equational Reasoning to prove (Q 3 2) evaluates to 6.

Expression	Justification
(Q 3 2)	Premise
(if (= 2 1) 3 (+ 3 (Q 3 (- 2 1))))	Apply definition of Q with a=3 and b=2
(if #f 3 (+ 3 (Q 3 (- 2 1))))	Evaluate = (equals)
(+ 3 (Q 3 (- 2 1)))	Evaluate if
(+ 3 (Q 3 1))	Evaluate – (subtraction)
(+ 3 (if (= 1 1) 3 (+ 3 (Q 3 (- 1 1)))))	Apply Definition of Q with a=3 and b=1
(+ 3 (if #t 3 (+ 3 (Q 3 (- 1 1)))))	Evaluate = (equals)
(+ 3 3)	Evaluate if
6	Evaluate +